

L13: Inclusion Probabilities with Each Sampling Strategy

An important question for each sampling strategy is: *what is the probability that a given sampling strategy includes a node or an edge in the sample?* As we will see later, these probabilities are essential in deriving important network estimators.

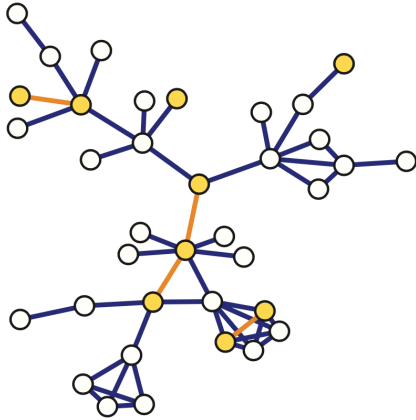
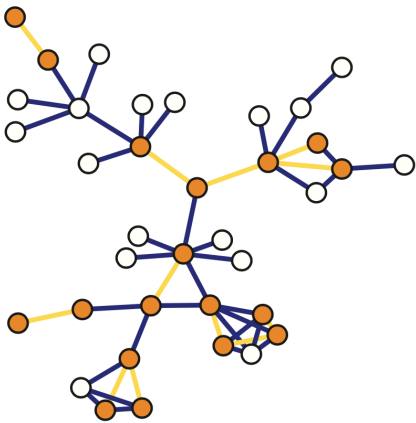


Image source: Kolaczyk, Eric D. *Statistical Analysis of Network Data: Methods and Models* (2009) Springer Science+Business Media LLC.

For **induced subgraph sampling** (see above), the node and edge inclusion probabilities are, respectively:

$$\pi_i = \frac{n}{N}, \quad \pi_{(i,j)} = \frac{n(n-1)}{N(N-1)}$$

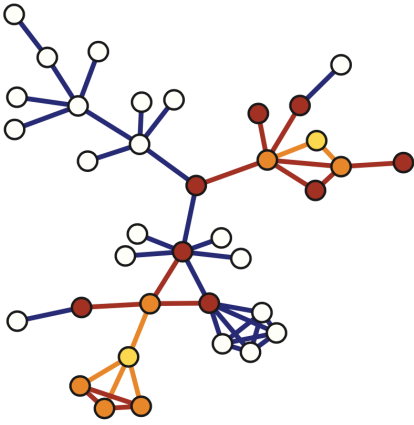
where n is the number of sampled nodes -- because each node of the population graph is sampled uniformly at random without replacement, and each edge of the population graph is sampled if the two corresponding nodes are sampled. Note that the node inclusion probability is the same for all nodes (and the same is true for the edge inclusion probability).



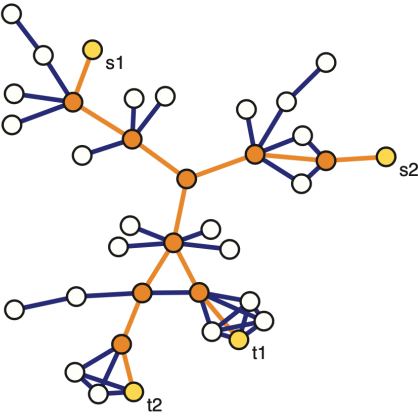
For **incident subgraph sampling** (see above), recall that we sample n edges randomly and without replacement. The inclusion probability for edges is simply: $\pi_{(i,j)} = n/|E|$, where $|E|$ is the number of edges in the population graph. The inclusion probability for node i is one minus the probability that none of the k_i edges of node i is sampled:

$$\pi_i = 1 - \frac{\binom{|E|-k_i}{n}}{\binom{|E|}{n}}, \quad \text{if } n \leq |E| - k_i$$

Of course if $n > |E| - k_i$, then $\pi_i = 1$. Note that, with incident subgraph sampling, nodes of higher degree have a higher inclusion probability.



With **snowball sampling** (see above -- the yellow nodes are the seeds and the process has two stages: first orange nodes and then brown nodes) , the inclusion probabilities are harder to derive, especially when the snowball process includes multiple stages. The statistical literature provides some approximate expressions, however.



With **link tracing** (including traceroute sampling -- see above), the inclusion probabilities are also harder to derive. Suppose however that the "traceroutes" (sampled paths from source nodes to target nodes) follow only **shortest-paths**, a fraction ρ_s of nodes is marked as sources while another fraction ρ_t of nodes is marked as targets. and b_i is the betweenness centrality of node i while $b_{(i,j)}$ is the betweenness centrality of edge (i,j) . Then, the node and edge inclusion probabilities are approximately:

$$\pi_i \approx 1 - (1 - \rho_s - \rho_t)e^{-\rho_s \rho_t b_i}, \quad \pi_{(i,j)} \approx 1 - e^{-\rho_s \rho_t b_{(i,j)}}$$

Food For Thought

Search the literature to find at least an approximate expression for the inclusion probability of nodes and edges in two-step snowball sampling.